

Specification

產品規格書

產品名稱 (Product): Wi-Fi 802.11 b/g/n 1x1 2.4GHz

解決方案 (Solution): Realtek RTL8711AM-VA1 QFN Package

產品型號 (Model No.):

DK-9169 Realtek RTL8711AM-VA1 Module Chip Antenna

Issued Date: 2022/01/01

1. Description

Ameba RTL8711AM is a highly integrated single-chip with low power consumption for IoT (Internet of Things). It combines an Arm® new architecture Arm®v8M KM3 MCU, Wi-Fi and provides a bunch of configurable GPIOs which are configured as digital peripherals for different applications and control usage.

RTL8711AM SoC also integrates internal 2.5MB RAM memory to minimize IoT end product size and reduce your development cost.

Features:

1. Complete IoT solution (low-level development, cloud services, APP)
2. Support AT + application set secondary development
3. Support OTA wireless upgrade
4. Support Bluetooth smart networking,

1.1. Application

- IoT
 - Home automation
 - Sensor networks
 - Building automation
 - Industrial
 - Retail
- Interactive entertainment devices
 - Remote control
 - Gaming controller

1.2. Features

- ARMv8M KM3, 166MHz
- 802.11 b/g/n 1×1 Wi-Fi (20/40MHz)
- Wi-Fi simple config
- 2.5 MB internal RAM on SoC and 4MB external flash on Dev. Board
- Compact size with (L) 25 x (W) 17 x (H) 2.9 mm.

2. Product Dimension

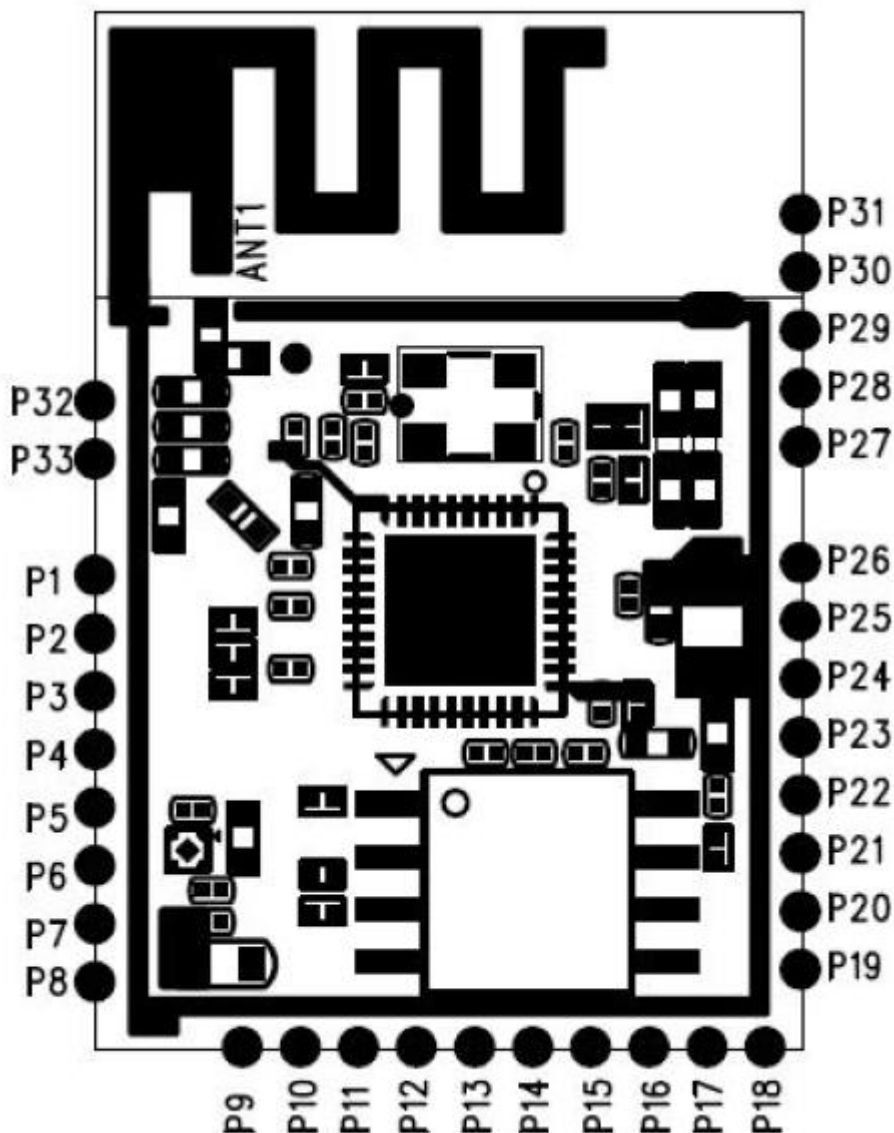
2.1. Pin Assignment

Pin No.	Name	Pin function	Description
1	GND	Ground	The pad must be connected to a solid ground plane
2	GND	Ground	The pad must be connected to a solid ground plane
3	ADC	Digital I/O	General purpose I/O / ADC_CH2
4	GPIOA_3	Digital I/O	General purpose I/O
5	GPIOA_5	Digital I/O	General purpose I/O
6	GPIOA_7	Digital I/O	General purpose I/O
7	GPIOA_6	Digital I/O	General purpose I/O
8	VDD3.3V	Power	3.3 V Power supply
9	GPIOC_5	Digital I/O	General purpose I/O
10	GPIOC_4	Digital I/O	General purpose I/O
11	GPIOC_2	Digital I/O	General purpose I/O
12	NC		
13	GPIOC_3	Digital I/O	General purpose I/O
14	GPIOC_1	Digital I/O	General purpose I/O
15	GPIOC_0	Digital I/O	General purpose I/O
16	NC		
17	NC		
18	NC		
19	NC		
20	NC		
21	NC		
22	NC		
23	NC		
24	CHIP_EN		Enable Chip 1:enable Chip ; 0:Shutdown Chip
25	NC		
26	NC		
27	JTAG_CLK	Digital I/O	
28	JTAG_TMS	Digital I/O	
29	JTAG_TDO	Digital I/O	
30	JTAG_TDI	Digital I/O	
31	JTAG_TRST	Digital I/O	
32	GPIOB_1	Digital I/O	General purpose I/O / UART 2_IN / UART_LOG_RX
33	GPIOB_0	Digital I/O	General purpose I/O / UART 2_OUT / UART_LOG_TX

2.2. PCB Dimensions & Pin Indication

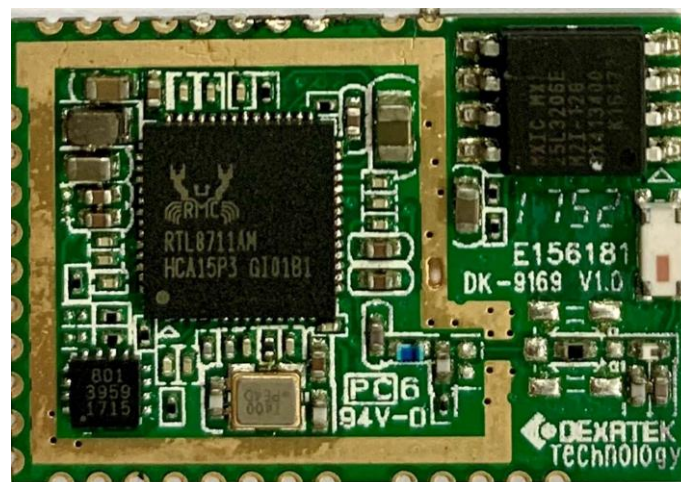
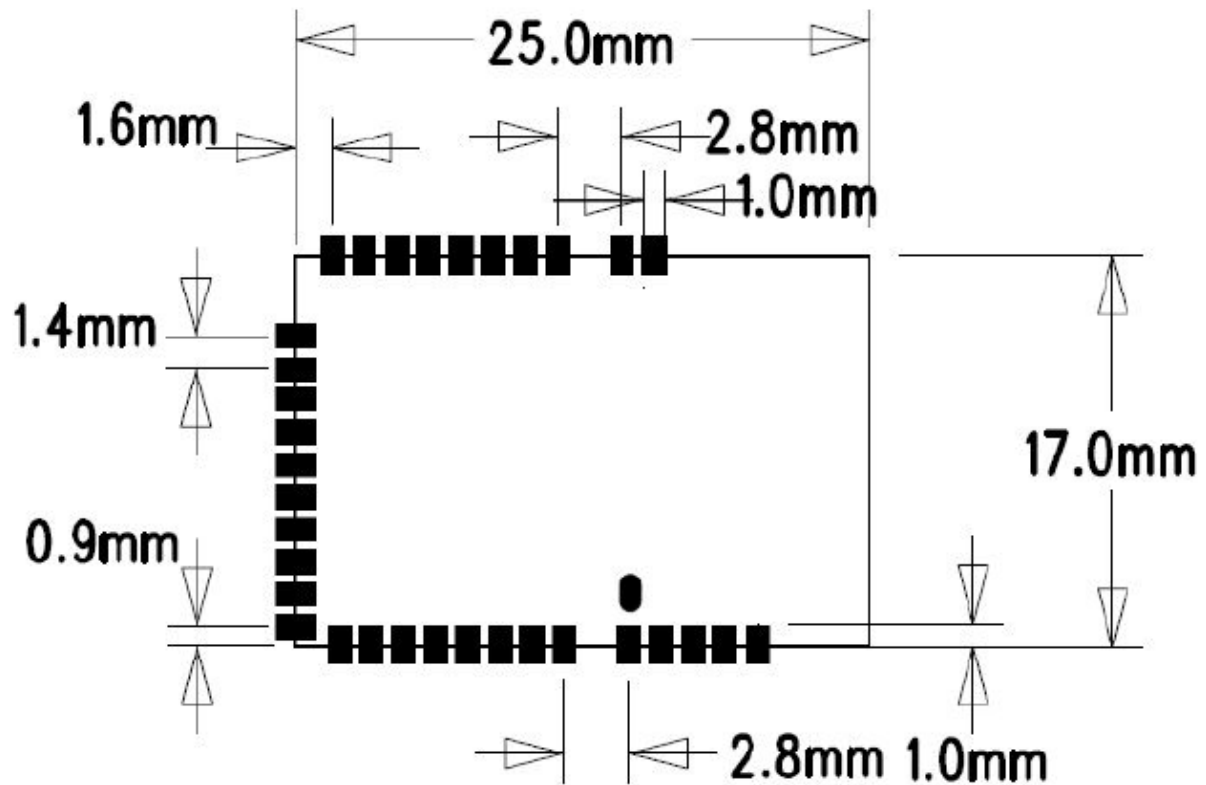
- DK-9169 Realtek RTL8711AM-VA1 Module PCB Antenna

	PCB Size (in mm)		
	Min.	Norm	MAX.
L		25	
W	-0.15	17	+0.2
H		2.9	



2.3. Recommended Layout of Solder Pad

Graphs are all in Top View, Unit in mm

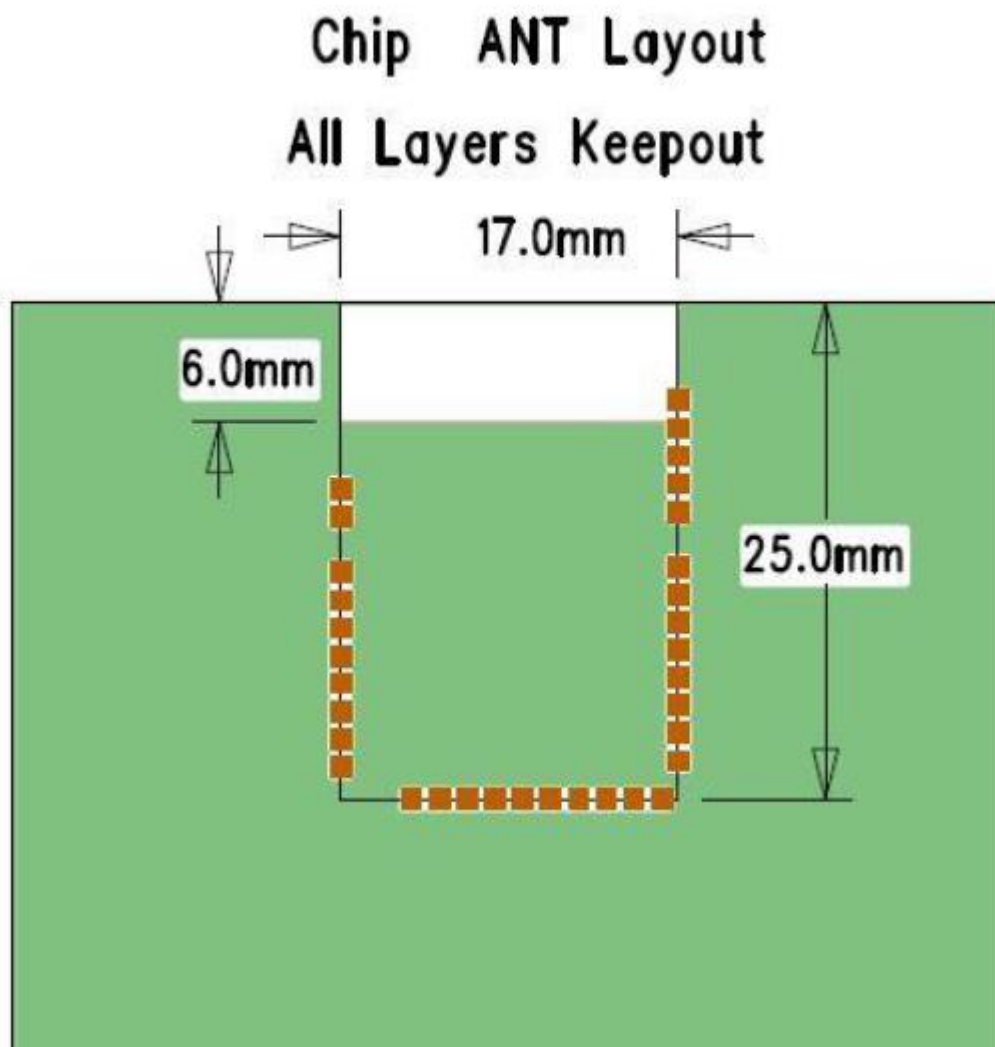


2.4. RF Layout Suggestion (aka Keep-Out Area)

Please follow below instruction to have better wireless performance. Make sure to keep the “No-Ground-Pad” as wider as you can when there is no enough space in your design.

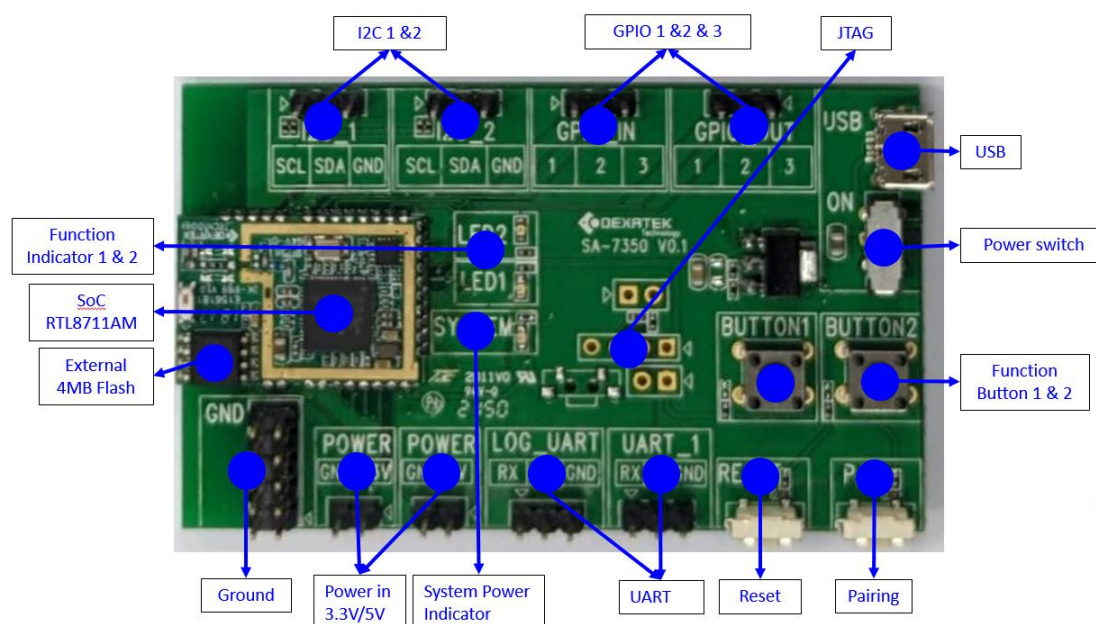
No Ground Pad should be included in the corresponding position of the antenna in **EACH LAYER**.

Place the module towards the edge of PCB to have better performance than placing it on the center.



DK-9169 EVB Guide

The DK-9169 EVB is based on the RTL8711AM SoC and designed for IoT System Extended



DK-9169 EVB Spec	
SoC	RTL8711AM
WLAN	WiFi 802.11 b/g/n 2.4
Antenna	Chip Antenna
USB	USB2.0 *1
RAM	2.5MB internal RAM on SoC & 4MB external flash on Dev. Board
Indicator	Function*2 (Yellow & Orange) System Power(Blue)
Button	Function Button *2
UART	UART *2
GPIO	GPIO IN*3 + OUT*3
I2C	I2C*2
Power	3.3V / 5V
EVB SIZE	75.69mm 44.83mm